Multiple Choice:

1. The mass of one Avogadro number of helium atom is ____.

A. 1 g
B. 4 g
C. 8 g
D. 4 x ($6.02x10^{23}$) g

2. What is the mass of an electron in amu?

A. 2.490210 x 10^-4
B. 3.890140 x 10^-4
C. 1.007276 x 10^-4
D. 5.485799 x 10^-4

3. The number of __________ in an atom defines the isotopes of an element.

A. neutrons
B. protons
C. electrons
D. protons and neutrons

4. What refers to the measure of the atom's tendency to attract an additional electron?

A. Period number
B. Electron affinity
C. Ionization energy
D. Electronegativity

5. Elements in Gropu1A in the periodic table are ______.

A. Boron group
B. Alkaline earth metals
C. Alkali metals
D. Carbon group

6. The boron group is what group in the periodic table?

A. Group IIIA
B. Group IVA
C. Group VA
D. Group VIA

7. The chalcogens are elements in what group in the periodic table?

A. Group VA
B. Group VIA
C. Group VIIA
D. Group IVA

8. Which of the following is NOT a noble gas?

A. Argon
B. Xenon
C. Radon
D. Antimony

9. The elements of groups IA, IIA, IIIA, IVA, VA, VIA, VIIA and VIIIA are called main group elements or

A. transition
B. representatives
C. inner transitions
D. metals or nonmetals

10. The most electronegative element among the following is:

A. Sodium
B. Bromine
C. Fluorine
D. Oxygen

11. The ionization energy is _________ to the atomic size of atom.

A. directly proportional
B. inversely proportional
C. equal to
D. not related

12. "When two or more elements form more than one compound, the ratio of the masses of one element that combine with a given mass of another element in the different compounds is the ratio of small whole numbers." This statement is known as:

A. Graham' Law of diffusion
B. The uncertainty principle
C. Law of definite proportion
D. Law of multiple proportion

13. "The total pressure of a mixture of gases equals the sum of the partial pressures of each of the gases in the mixture". This statement is known as __________.

A. Dalton's law of partial pressure
B. Gay-Lusaac law
C. Boyle's law
D. Charle's law

14. What law states that the rate of effusion of a gas, which is the amount of gas that through the hole in a given amount of time, is inversely proportional to the square root of its molar mass?

A. Henry's law
B. Graham's law of effusion
C. Hund's law
D. Lewis theory

15. The law which states that the amount of gas dissolved in a liquid is proportional to its partial pressure.

A. Dalton's Law
B. Gay Lussac's Law
C. Henry's Law
D. Raoult's Law

16. A motorist starts a trip on a cold morning when the temperature is 4 degrees Celsius and she checks her tire pressure at a gas station and finds the pressure gauge reads 32 psi. After driving all day, her tires heat up and by afternoon the tire temperature has risen to 50 degrees Celsius. Assuming the tire volume is constant, to what pressure will the air in the tires have risen?

A. 54.8 psig
B. 54.8 psia
C. 39.8 psia
D. 49.8 psig

17. What is the ratio of the number of moles of solute per kilogram of solvent?

A. Molarity
B. Molality
C. Formality
D. Mole fraction

18. Calculate the volume at STP occupied by 12.5 g of nitrogen gas.

A. 4.5 L
B. 6.7 L
C. 8 L
D. 10 L

19. How many grams of oxygen are there in 2.5 moles of $Ca(NO_3)_2$?

A. 120 g C. 200 g
B. 160 g D. 240 g

20. Calculate the mass at STP, in grams, of 230 ml of Cl_2 gas at 20°C and 740 torr pressure.

A. 0.444 g C. 0.662 g
B. 0.581 g D. 0.714 g

21. What is the density of propane gas (C_3H_8) if the pressure of the gas was measured to be 3.5 atm at 285 K?

A. 3.2 g/L C. 7.8 g/L
B. 6.6 g/L D. 10.0 g/L

22. What would be the number of moles of chlorine gas present if the gas fills a 75-ml flask having a pressure of 755 mmHg and a temperature of 100°C?

A. 0.00112 C. 0.00243
B. 0.00165 D. 0.00411

23. If a gas diffuses at a rate of ½ as fast as O_2, what is the molecular weight of the gas?

A. 128 g/mol C. 182 g/mol
B. 166 g/mol D. 256 g/mol

24. What is the total pressure inside a container that contains He at a pressure of 0.45 atm and Ar at a pressure of 0.75 atm?

A. 0.8 atm C. 1.2 atm
B. 1.0 atm D. 1.5 atm

25. A volume of 50 liters is filled with helium at 15°C to a pressure of 100 standard atmospheres. Assuming that the ideal gas equation is still approximately true at this high pressure, calculate approximately the mass of helium required.

A. 815 g C. 860 g
B. 840 g D. 920 g

26. Calculate the mole fraction of phosphoric acid in 25% aqueous phosphoric acid solution.

A. 0.047 C. 0.065
B. 0.058 D. 0.078

27. 2.5g of glucose is dissolved into 375 ml of water. What is the molarity of the solution?

A. 0.012M C. 0.067M
B. 0.037M D. 0.081M

28. Calculate the molarity obtained by dissolving 102.9 g H_3PO_4 into 642 ml final volume of solution.

A. 1.32M C. 1.64M
B. 1.44M D. 1.88M

29. How many grams of sodium chloride must you add to 750g of water to make a 0.35 molal solution?

A. 7.59 g C. 8.89 g
B. 8.11 g D. 9.42 g

30. Calculate the molar concentration of a 45% aqueous calcium chloride solution whose density is 1.10 g/ml?

A. 3.14M C. 4.12M
B. 3.38M D. 4.46M

31. Determine the pH given the following value: $[H^+] = 1x10^{-13}$

A. 1 C. 12
B. 5 D. 13

32. Determine the empirical formula of a compound with a percent composition of 40% sulfur and 60% oxygen.

A. S2O3 C. SO3
B. SO D. SO4

33. What is the molecular formula of a compound which has a gram formula mass of 78 g/mol and the empirical formula NaO.

A. NaO C. Na2O3
B. Na2O2 D. Na3O3

34. Ammonium sulfate has a chemical formula of $(NH_4)_2SO_4$. Calculate how many percent of this compound is oxygen?

A. 48.43% C. 43.34%
B. 49.43% D. 42.34%

35. A compound contains, by mass, 40.0% carbon, 6.71% hydrogen, and 53.3% oxygen. A 0.320 mole sample of this compound weighs 28.8 g. The molecular formula of this compound is:

A. $C_2H_4O_2$ C. C_2H_4O
B. $C_3H_6O_3$ D. $C_4H_7O_2$

36. A sample of nitrogen occupies 5.50 liters under a pressure of 900 torr at 25°C. At what temperature will it occupy 10.0 liters at the same pressure?

A. 45.45 °C C. 174.21 °C
B. 110.00 °C D. 268.94 °C

37. Pitchblende is the most important compound of uranium. Mass analysis of an 84.2-g sample shows that it contains 71.4 g of uranium, with oxygen the only other element. How many grams of uranium are in 102 kg of pitchblende?

A. $7.13x10^4$ g C. $8.36x10^4$ g
B. $8.03x10^4$ g D. $8.65x10^4$ g

38. What happens to the pressure of a sample of helium gas if the temperature is increased from 200 K to 800 K, with no change in volume?

A. P will decrease by a factor of 4
B. P will increase by a factor of 4
C. P will not change
D. P will increase by a factor of 2

39. What happens to the pressure of a sample of helium gas if the volume is reduced from 6 liters to 3 liters, with no change in temperature?

A. P will decrease by a factor of 4
B. P will increase by a factor of 4
C. P will not change

D. P will increase by a factor of 2

40. Which of the following is NOT a property of acids?

A. Taste sour
B. Feel slippery on the skin
C. Turn litmus paper to red
D. Dissolve metals producing various salts and hydrogen gas

41. What is the atomic number of oxygen?

A. 6
B. 7
C. 8
D. 16

42. What refers to the passage of molecules of a gas from one container to another through a tiny opening between the containers?

A. Diffusion
B. Effusion
C. Fission
D. Fusion

43. What is the heaviest subatomic particle?

A. Electron
B. Proton
C. Neutron
D. Atom

44. The properties of a material that do not change when the amount of substance changes are called _______ properties.

A. Intensive
B. extensive
C. chemical
D. physical

45. "When the elements are arranged in the order of increasing atomic number, elements with similar properties appear at periodic intervals." This statement is known as _________.

A. Law of multiple proportion
B. Law of definite proportion
C. The periodic law
D. Dalton's atomic theory

46. What happens to the atomic size of the elements in a group when you go from top to bottom of the group?

A. It remains the same
B. It increases
C. It decreases
D. It becomes zero

47. What refers to the passage of molecules of a gas from one container to another through a tiny opening between the containers?

A. Diffusion
B. Effusion
C. Fission
D. Fusion

48. Calculate the percentage composition of hydrogen in sodium hydroxide (NaOH). 1 mole of NaOH weighs 40.0 g.

A. 8.5 %   C. 57.5 %
B. 40.0 %   D. 2.5 %

49. Calculate the oxygen pressure in a mixture of 0.500 mol of oxygen and 0.750 mol of nitrogen with a total pressure of 40.0 kPa.

A. 15.0 kPa   C. 16.0 kPa
B. 15.5 kPa   D. 16.5 kPa

50. In a certain experiment, argon effuses from a porous cup at 4.00 mmol/minute. How fast (in mmol/min) would chlorine effuse under the same condition?

A. 3.00   C. 4.00
B. 3.50   D. 4.50